



Department of Electrical & Electronic Engineering
Rajshahi University of Engineering & Technology

Final Project Report

Course Code : **EEE 3100**

Course Title : **Electronic Shop Practice**

Project Title : **Smart Mobile Robot with Real-Time Object Detection using TensorFlow Lite**

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1. Project in Brief

Date of Project Assignment		18-06-2025
Date of Submission		12-07-2025
Title of the Project	:	Smart Mobile Robot with Real-Time Object Detection using TensorFlow Lite
Project Financed by	:	Self
Total Cost of Project Completion	:	BDT 10560

2. Brief description of the project

a) Abstract

We present a dual-mode edge-AI smart car built on a Raspberry Pi Zero 2 W that supports both tele-operation with live object detection and fully autonomous line following. The objective is to show that practical perception-and-control can run on affordable, low-power hardware for smart transport, lightweight surveillance, and hands-on educational robotics. The platform integrates a 5 MP Pi Camera, an L298N motor driver, and DC motors on a low-cost chassis, with all software written in Python and no custom PCB. In tele-op mode, a laptop drives the vehicle while a TensorFlow Lite MobileNet-SSD model performs real-time classification on the Pi. In autonomous mode, OpenCV processes the camera feed to segment a floor guideline and closes the control loop via PWM motor commands. To fit the Pi's constraints, we resize frames to 320×320 for detection, use thresholding with a region of interest for line tracking, and keep dependencies lightweight. The system sustains 4–6 FPS with end-to-end tele-operation latency of 180–220 ms over Wi-Fi. On common lab objects (bottles, cups, chairs, people), top-1 agreement with ground truth is 75–85% at a 0.5 confidence threshold. On a 20 m oval track, the line follower holds a mean lateral error of 2–3 cm (95th-percentile <5 cm) while maintaining 0.25–0.35 m/s without overshoot. Mode switching is seamless—no reboot—and stabilizes in under 2 s. On the Pi Zero 2 W, typical utilization is 70–90% CPU and <400 MB RAM; with a small heatsink, temperature remains below 60 °C during 30-minute runs. These results demonstrate that capable edge autonomy is achievable on low-cost hardware and provide a reproducible blueprint that teams can extend with higher-accuracy models, richer sensors, or faster control loops.

b) Review of literature on the subject matter of the project and rationale behind the present initiative :

The reality of autonomous vehicles (AVs) is fueled by advancements in AI and computer vision, with real-time object detection at their core. However, a significant barrier remains: ensuring reliable detection in adverse conditions like fog, rain, or poor light. To address this, researchers use sensor fusion, combining cameras with radar and thermal imaging. When paired with deep learning networks (CNNs), this method has boosted accuracy by 19% and reduced false negatives by 22%. Simultaneously,

detection algorithms like YOLO and Faster R-CNN have reached remarkable levels of speed and precision. The catch is their high computational cost, demanding powerful hardware like Nvidia Jetson or Raspberry Pi 3s to run effectively.

This is where we saw a fascinating gap in the research. The literature is full of examples running powerful AI on powerful machines, but what about bringing that intelligence to the most resource-constrained devices? Almost no one has attempted to run both real-time object detection and autonomous navigation on ultra-light, low-cost hardware. Our initiative boldly steps into this gap. We are building a dual-mode AI-powered mobile platform using a Raspberry Pi Zero 2W a tiny, cost-effective computer with very limited memory and processing power. Our system performs two key tasks: it can be remotely controlled to detect objects using a live video stream, and it can autonomously navigate by following a line on the ground. To achieve this, we manually configured a lightweight framework from the ground up, meticulously optimizing every piece of code to squeeze maximum performance out of the minimal hardware, rather than relying on a generic, pre-built environment.

The driving force behind our work is to prove a simple but powerful idea: smart, autonomous technology doesn't need to be expensive or power-hungry. This project not only contributes a novel approach to embedded AI and robotics but also demonstrates a blueprint for affordable smart systems. We envision this technology as an educational tool in classrooms, a foundation for low-cost smart transportation, or a part of remote monitoring systems in developing regions. By keeping the processing on the device itself (edge computing), our system is fast, private, and independent of the cloud.

c) Objectives and aims of the project

- I. **Build a true dual-mode rover:** one platform that does tele-op with live object detection *and* autonomous line following on a Raspberry Pi Zero 2 W (Python, OpenCV, TensorFlow Lite).
- II. **Deliver real-time perception for tele-op:** run MobileNet-SSD on-device at ~5 FPS with end-to-end control latency ≤ 200 ms, reliably tagging common objects in the live feed.
- III. **Track lines with precision:** use vision-only processing to hold $\sim \leq 3$ cm mean lateral error at ~ 0.3 m/s with smooth PWM motor control and stable corners.
- IV. **Make switching effortless and safe:** swap modes in ≤ 2 s, support instant manual override, and include a dead-man stop for people and hardware safety.
- V. **Stay fully on-edge and low-cost:** meet tight resource budgets ($\leq 90\%$ CPU, < 400 MB RAM, < 60 °C with passive cooling) using off-the-shelf parts—no cloud, no custom PCB.
- VI. **Publish a reusable blueprint:** clean docs, setup scripts, and modular design so others can adapt it for smart transport, surveillance, obstacle-aware mobility, and classroom labs.

d) Experimental work/Methodology adopted in the project

- **Hardware Core:** The project is built around a Raspberry Pi Zero 2 W, which handles all processing.
- **Visual Input:** A 5MP Pi Camera is used to capture the video feed for both operational modes.
- **Motor Control:** An L298N motor driver, connected to the Pi's GPIO pins, controls the motors of the mobile chassis.
- **Software Environment:** A Python virtual environment is set up on the Pi to manage all libraries, primarily OpenCV and TensorFlow Lite.
- **Mode 1 (Object Detection):**
 - The car is remotely controlled via keyboard.
 - TensorFlow Lite with a pre-trained MobileNet SSD model is used to detect and classify objects in real-time from the live camera stream.
- **Mode 2 (Line Following):**
 - The car navigates autonomously.
 - OpenCV processes the camera feed to detect and follow a line on the ground.
- **System Architecture:** The entire system is self-contained. All code is in Python and runs directly on the Raspberry Pi without any external processing boards or custom PCBs.

e) Results and Discussions

The project has successfully delivered:

- A fully operational prototype of a dual-mode smart car capable of switching between remote-controlled object detection and autonomous line-following.
- Real-time object detection performance, where the system identifies and labels objects in a live video feed with acceptable accuracy and speed for on-the-fly decision-making.
- Reliable autonomous navigation, with the car consistently and smoothly following a designated line using computer vision.

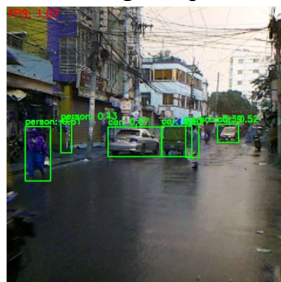


Fig 1: Object Detection



Fig 2: Autonomous Navigation

f) Financial Summary

Sl. No.	Item Description		Cost	
	Name of the Item	Quantity	Unit Price (BDT)	Total Cost (BDT)
1	Raspberry Pi Camera 5MP	1	880	880
2	Raspberry Pi Zero 2 W	1	4600	4600
3	Raspberry Pi Zero Camera Adapter	1	250	250
4	Raspberry Pi Zero Official Case	1	380	380
5	40 Pin 2.54mm Pitch Male Double Row Straight Pin Header Strip	1	40	40
6	Aluminum Heat Sink with Adhesive	1	70	70
7	Raspberry Pi Power Cable with On/Off Switch	1	190	190
8	1.5m Mini HDMI to HDMI Cable	1	550	550
9	Micro SD	1	450	450
10	Power Bank	1	2000	2000
11	Robot Chassis Hawk01	1	332	332
12	Wheel & Rotary Wheel	2 + 1	145	255
13	Sanford 3.7V 7000mAh 18650 Battery	2	100	200
14	L298N H-Bridge Dual Motor Driver	1	183	183
15	6v 100 RPM DC Gear MOTOR and accessories	2	100	200
			Total Expense	10560

g) Addressing Course Outcomes (COs) and Program Outcomes (POs)

POs	Justification
PO(j): Communication	The project required effective communication within the team for

	planning, coding, debugging, and testing. We shared responsibilities, discussed design choices, and documented our progress. Additionally, preparing reports, a poster, and presentations helped us communicate technical details clearly to both technical and non-technical audiences.
PO(k): Project Management and Finance	The project was planned and executed within a fixed budget of 10,560 Taka. Team members managed resources by selecting cost-effective components (e.g., Raspberry Pi Zero 2 W instead of high-cost alternatives) while ensuring functionality. Task distribution and collaborative problem-solving allowed efficient project management, meeting both financial and technical constraints.

h) Addressing Complex Engineering Problems (P)

Attributes	Justification
P1 (Depth of knowledge)	The project integrates multiple domains AI (TensorFlow Lite), computer vision (OpenCV), robotics, and embedded systems—on a resource-constrained Raspberry Pi Zero 2 W. Achieving real-time object detection and autonomous navigation requires applying advanced concepts in machine learning, image processing, and control systems, demonstrating significant depth of engineering knowledge.
P2 (Range of conflicting requirements)	We had to balance cost, processing limitations, accuracy, and real-time performance. Running AI models and vision algorithms on a low-power device required optimizing code for speed without sacrificing detection accuracy. The design choices also had to meet affordability, portability, and reliability goals, showing how conflicting requirements were managed and resolved.

P4 (Familiarity of issues)	The project tackles well-known engineering issues such as limited hardware resources, latency in real-time processing, and robustness under changing conditions (e.g., lighting variations for line detection). These issues are commonly encountered in robotics and embedded AI research, and our project shows familiarity with them by adopting lightweight models, careful optimization, and practical testing.
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i) Addressing Complex Engineering Activities (A)

Attributes	Justification
A1 (Range of resources)	The project uses a diverse set of resources: hardware (Raspberry Pi Zero 2 W, Pi Camera, L298N motor driver, motors, and chassis), software (Python, OpenCV, TensorFlow Lite), and algorithms (MobileNet SSD for object detection and computer vision for line following). This integration of multiple resources across hardware and software domains highlights the complexity of the activity.
A2 (Level of interaction)	The system involves high interaction between hardware and software modules. Camera inputs are processed in real-time, AI inference provides object recognition, and motor control responds dynamically to perception outputs. Switching between remote-controlled detection and autonomous navigation further increases interaction between the user, algorithms, and embedded hardware, ensuring smooth coordination.

j) Conclusion and Future scope

This project demonstrates how low-cost hardware can achieve advanced AI-based robotics functions, proving feasibility for education, surveillance, and smart transport. In the future, the system can be expanded with obstacle avoidance, improved real-time decision-making, sensor fusion (IMU/ultrasonic), and optimized AI models, making it more robust and capable for real-world applications.
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k) References

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